Exercise list R2c — Financial planning and forecasting for valuation

Syllabus reading "Valuation and financial modeling: a case study" (Berk & DeMarzo, *Corporate Finance*, Ch. 19; the local *Fundamentals* edition has it as Ch. 18, "Financial Modeling and Pro Forma Analysis"): pro forma statements by the percent-of-sales method, external financing required (EFN, "net new financing", "NAF") and the plug, forecasting a planned expansion with lumpy capacity, internal and sustainable growth, growth and firm value, valuing an expansion from forecast free cash flows, and sensitivity/scenario analysis. 15 real exercises ordered easy → really hard, copied verbatim from the course books for personal study (Ross statements stay in Portuguese); pages are printed, with the PDF page in brackets. None of them appears in another list (Higgins 3.13–3.14 are in R5b). Links: exercise index · course program.

Block	Exercises
Percent-of-sales pro forma, the plug, EFN	1, 2, 3, 5, 6, 7, 8
Capacity, EFN with a financing policy (constant D/E)	11, 14
Internal and sustainable growth, growth vs. value	4, 9, 10, 13, 14
Sensitivity and scenario analysis	8 (payout cap), 10 (d), 12
Planned expansion, free cash flows, continuation value, APV	15

Coverage note: every subtopic has book exercises. The one piece of the chapter with no exercise of its own is the *optimal timing* of the KMS expansion (Berk §18.5, "Optimal Timing and the Option to Delay"); exercise 15 values the expansion under a slower-growth scenario that delays it to 2012, which is the closest the books come.

References

#	Tier	Book	Ch.	Exercise	Printed p. (PDF)	Answer in book	Trains
1	Easy	Berk–DeMarzo–Harford, *Fundamentals* 2e	Ch. 18	18.1	558 (589)	no	One line of the percent-of-sales method: cost of goods sold as a fixed fraction of sales
2	Easy	Higgins (Koski–Higgins–Mitton) 13e	Ch. 3	3.1 + 3.4	103 (124)	3.1 yes, p. 405; 3.4 no	Reading the sign of EFN; what raises EFN; annual forecast vs. intra-year needs
3	Easy	Ross *Fundamentos* 9e	Cap. 4	4.1 + 4.2	117 (141)	no	The plug variable (dividends, then EFN) in a two-line pro forma

#	Tier	Book	Ch.	Exercise	Printed p. (PDF)	Answer in book	Trains
4	Easy	Berk–DeMarzo–Harford, *Fundamentals* 2e	Ch. 18	18.16	561 (592)	no	Internal growth = ROA × b, sustainable growth = ROE × b, effect of payout
5	Easy	Higgins (Koski–Higgins–Mitton) 13e	Ch. 3	3.5	104 (125)	yes, p. 405	Finishing a pro forma: fixed-asset roll-forward, equity roll-forward, EFN as the plug
6	Exam	Berk–DeMarzo–Harford, *Fundamentals* 2e	Ch. 18	18.2 + 18.3	558 (589)	no	Net new financing from projected assets and sources; splitting it to hold D/E constant
7	Exam	Ross *Fundamentos* 9e	Cap. 4	4.9–4.11	118–119 (142–143)	no	Full percent-of-sales balance sheet with spontaneous payables; negative EFN
8	Exam	Berk–DeMarzo–Harford, *Fundamentals* 2e	Ch. 18	18.8–18.11	560 (591)	no	Two-pass pro forma (KMS method), net new financing, payout policy that caps external financing

#	Tier	Book	Ch.	Exercise	Printed p. (PDF)	Answer in book	Trains
9	Exam	Berk–DeMarzo–Harford, *Fundamentals* 2e	Ch. 18	18.17–18.19	561 (592)	no	KMS growth vs. its SGR; new debt at the SGR; D/E drift at the internal growth rate
10	Exam	Higgins (Koski–Higgins–Mitton) 13e	Ch. 4	4.8	135–136 (156–157)	no	$g^* = PRA\hat{T}$ over five years; sensitivity of g^* to margin and leverage
11	Exam	Ross *Fundamentos* 9e	Cap. 4	4.25–4.27	121 (145)	no	EFN with fixed interest, excess capacity, and a constant debt-equity policy
12	Really hard	Higgins (Koski–Higgins–Mitton) 13e	Ch. 3	3.9 (with 3.8 as Part 0)	105 (126)	3.9 yes, p. 406; 3.8 no	Two-year spreadsheet forecast with circular interest; sensitivity and recession scenario
13	Really hard	Ross *Fundamentos* 9e	Cap. 4	4.31–4.33 (DESAFIO)	122 (146)	no	Deriving EFN(g), the internal and sustainable growth formulas, beginning- vs. end-of-period ROE

#	Tier	Book	Ch.	Exercise	Printed p. (PDF)	Answer in book	Trains
14	Really hard	Ross *Fundamentos* 9e	Cap. 4	4.28 + 4.29 (DESAFIO; 4.25, 4.27 as Part 0)	121 (145)	no	EFN as a function of g; growth rates that zero EFN vs. the textbook formulas, and why they differ
15	Really hard	Berk–DeMarzo–Harford, *Fundamentals* 2e	Ch. 18	18.20–18.22 (18.12–18.15 as Part 0)	561 (592)	no	KMS under 0.25%/yr share growth: capacity, delayed financing, net income, NWC, FCF, EBITDA-multiple continuation value, APV

Tier 1 — Easy

1. Cost of goods sold by the percent of sales method

Source: Berk, Ch. 18, Exercise 18.1, p. 558 (PDF 589) · Answer in book: no

1. Your company has sales of $100,000 this year and cost of goods sold of $72,000. You forecast sales to increase to $110,000 next year. Using the percent of sales method, forecast next year's cost of goods sold.

---

2. Reading external financing required

Source: Higgins, Ch. 3, Exercises 3.1 and 3.4, p. 103 (PDF 124) · Answer in book: yes for 3.1, p. 405; no for 3.4

1. Suppose you constructed a pro forma balance sheet for a company and the estimate for external financing required was negative. How would you interpret this result?

2. Answer true or false to each of the following. Briefly explain your reasoning for each answer.

a. All else equal, increasing the projected amount of accounts receivable in a financial forecast will increase external funding required.

b. Estimates of external funding required based on cash flow forecasts are usually higher than estimates based on pro forma financial statements.

c. An annual financial forecast for 2022 showing no external funding required assures a company that no cash shortfalls are likely to occur during 2022.

---

3. Corporação Felipe: the plug variable

Source: Ross, Cap. 4, Exercises 4.1 and 4.2, p. 117 (PDF 141) · Answer in book: no (selected answers in the web-only Apêndice C)

1. **Demonstrações projetadas** [OA1] Considere as seguintes demonstrações contábeis simplificadas da Corporação Felipe (assumindo que não há imposto de renda):

Demonstração de resultados		Balanço patrimonial			
Vendas	\$23.000	Ativo	\$15.800	Dívida	\$ 5.200
Custos	16.700			Patrimônio líquido	10.600
Lucro líquido	\$ 6.300	Total	\$15.800	Total	\$15.800

A Felipe previu um aumento de 15% nas vendas. Também previu que cada item do balanço patrimonial crescerá 15%. Construa as demonstrações projetadas e deixe-as equilibradas. Qual é a variável de fechamento?

2. **Demonstrações projetadas e NAF** [OA1, OA2] Na questão anterior, assuma que a Felipe distribua metade do lucro líquido como dividendos. Os custos e o ativo variam com as vendas, mas a dívida e o patrimônio líquido não. Prepare as demonstrações projetadas e determine o aporte financeiro necessário.

---

4. Internal and sustainable growth rates

Source: Berk, Ch. 18, Exercise 18.16, p. 561 (PDF 592) · Answer in book: no

16. Using the information in the table below, calculate this company's:

Net Income	50,000
Beginning Total Assets	400,000
Beginning Stockholders' Equity	250,000
Payout Ratio	0%

a. Internal growth rate.

b. Sustainable growth rate.

c. Sustainable growth rate if it pays out 40% of its net income as a dividend.

Context needed to solve it *From Berk §18.4, "Sustainable Growth Rate and External Financing", p. 548 (PDF 579):*

Internal Growth Rate Formula. Both these benchmark growth rates are aimed at identifying how much growth a firm can support based on its existing net income. For a firm that does not pay any dividends, its internal growth rate is its return on assets, because that tells us how fast it could grow its assets using only its net income. If the firm pays some of its net income out as a dividend, then its internal growth rate is reduced to only the growth supported by its retained earnings. This reasoning suggests a more general formula for the internal growth rate:

$$\text{Internal Growth Rate} = \left(\frac{\text{Net Income}}{\text{Beginning Assets}}\right) \times (1 - \text{Payout Ratio}) = \text{ROA} \times \text{Retention Rate} \qquad (18.4)$$

[…] **Sustainable Growth Rate Formula.** The sustainable growth rate allows for some external financing. It assumes that no new equity will be issued and that the firm's managers want to maintain the same debt-to-equity ratio. Thus, it tells us how fast the firm can grow by reinvesting its retained earnings and issuing only as much new debt as can be supported by those retained earnings. The formula for the sustainable growth rate is:

$$\text{Sustainable Growth Rate} = \left(\frac{\text{Net Income}}{\text{Beginning Equity}}\right) \times (1 - \text{Payout Ratio}) = \text{ROE} \times \text{Retention Rate} \qquad (18.5)$$

---

5. Universal Products: finishing a pro forma

Source: Higgins, Ch. 3, Exercise 3.5, p. 104 (PDF 125) · Answer in book: yes, p. 405

5. Universal Products Co. has almost finished its pro forma financial statements for 2022, as shown next.

Universal Products Co. — Financial Statements, 2021 and Pro Forma 2022 ($ thousands)

INCOME STATEMENT	Actual 2021	Forecast 2022	BALANCE SHEET	Actual 2021	Forecast 2022
Sales	$5,000	$5,500	Current assets	$2,500	$2,750
Cost of goods sold	4,000	4,400	Net fixed assets	2,000	_____
Operating expense	500	550	Total assets	$4,500	
Depreciation expense	250	260			
Interest expense	50	50	Current liabilities	$1,050	$1,155
Earnings before tax	200	240	Long-term debt	500	500
Tax	70	84	Owners' equity	2,950	_____
Net income	$130	$156	Total liabilities & equity	$4,500	

a. Assume that Universal plans to purchase $500,000 in fixed assets during 2022 and to dispose of no fixed assets during 2022. What would be its forecast for net fixed assets in 2022?

b. Assume that Universal plans to have a dividend payout ratio of 50 percent in 2022 and will neither sell nor repurchase equity during 2022. What would be its forecast for owners' equity in 2022?

c. Given the assumptions in questions (a) and (b), what is Universal's projected external funding required for 2022?

Tier 2 — Exam level

6. Net new financing and how to split it

Source: Berk, Ch. 18, Exercises 18.2 and 18.3, p. 558 (PDF 589) · Answer in book: no

2. For the next fiscal year, you forecast net income of $50,000 and ending assets of $500,000. Your firm's payout ratio is 10%. Your beginning stockholders' equity is $300,000 and your beginning total liabilities are $120,000. Your non-debt liabilities such as accounts payable are forecasted to increase by $10,000. What is your net new financing needed for next year?

3. Assume your beginning debt in Problem 2 is $100,000. What amount of equity and what amount of debt would you need to issue to cover the net new financing in order to keep your debt-equity ratio constant?

Context needed to solve it *From Berk §18.2, "Pro Forma Balance Sheet", p. 539 (PDF 570):*

The imbalance indicates that we will need $8396 in *net new financing* to fund our growth. **Net new financing** is the amount of additional external financing we will need to secure to pay for the planned increase in assets. It can be computed as:

$$\text{Net New Financing} = \text{Projected Assets} - \text{Projected Liabilities and Equity}$$

---

7. Companhia João Herdeiro: percent-of-sales balance sheet and NAF

Source: Ross, Cap. 4, Exercises 4.9–4.11, pp. 118–119 (PDF 142–143) · Answer in book: no (selected answers in the web-only Apêndice C)

9. **Cálculo dos lucros retidos a partir da demonstração financeira projetada** [OA1] Considere a seguinte demonstração de resultados da Companhia João Herdeiro:

COMPANHIA JOÃO HERDEIRO — Demonstração de resultados

Vendas		$38.000
Custos		18.400
Lucro tributável		$19.600
Impostos (34%)		6.664
Lucro líquido		$12.936
Dividendos	$5.200	
Acréscimo aos lucros retidos	7.736	

A taxa de crescimento projetada para as vendas é de 20%. Prepare a demonstração de resultados projetada assumindo que os custos variam com as vendas e a taxa de distribuição de dividendos é constante. Qual é a previsão de acréscimo aos lucros retidos?

10. **Aplicação da porcentagem de vendas** [OA1] A seguir está o balanço patrimonial da Companhia João Herdeiro. Com base nestas informações e na demonstração de resultados do problema anterior, forneça as informações que faltam usando a abordagem da porcentagem de vendas. Assuma que as contas a pagar variam com as vendas, mas não os empréstimos a pagar. Coloque "n/a" onde for necessário.

COMPANHIA JOÃO HERDEIRO — Balanço patrimonial

Ativo	$	Porcentagem de vendas	Passivo e patrimônio líquido	$
Ativo circulante			*Passivo circulante*	
Caixa	$ 3.050	—	Contas a pagar	$ 1.300
Contas a receber	6.900	—	Empréstimos	6.800
Estoque	7.600	—	Total	$ 8.100
Total	$17.550	—	Passivo não circulante	$25.000
Ativo imobilizado			*Patrimônio líquido*	
Instalações e equipamentos	$34.500	—	Capital social e ágio recebido na emissão de ações	$15.000
			Lucros retidos	3.950
			Total	$18.950
Total do ativo	$52.050	—	Total do passivo e do patrimônio líquido	$52.050

11. **NAF e vendas** [OA2] Com as duas questões anteriores, prepare um balanço patrimonial projetado mostrando a NAF. Assuma um aumento de 15% nas vendas, nenhuma dívida externa nova ou venda de novas ações e uma taxa de distribuição de dividendos constante.

---

8. Global Corp.: two-pass pro forma and a cap on new financing

Source: Berk, Ch. 18, Exercises 18.8–18.11, p. 560 (PDF 591) · Answer in book: no

For Problems 8–11, use the following income statement and balance sheet for Global Corp.:

Figures in $ millions

Income statement		Balance sheet	
Net sales	186.7	*Assets*	
Costs Except Depreciation	−175.1	Cash	23.2
EBITDA	11.6	Accounts Receivable	18.5
Depreciation and Amortization	−1.2	Inventories	15.3
EBIT	10.4	Total Current Assets	57
Interest Income (expense)	−7.7		
Pretax Income	2.7	Net Property, Plant, and Equipment	113.1
Taxes	−0.7	Total Assets	170.1

Income statement		Balance sheet	
Net Income	2.0		
		Liabilities and Equity	
		Accounts Payable	34.7
		Long-Term Debt	113.2
		Total Liabilities	147.9
		Total Stockholders' Equity	22.2
		Total Liabilities and Equity	170.1

8. Global expects sales to grow by 8% next year. Using the percent of sales method, forecast:

a. Costs except depreciation

b. Depreciation

c. Net income

d. Cash

e. Accounts receivable

f. Inventory

g. Property, plant, and equipment

h. Accounts payable

9. Assume that Global pays out 50% of its net income. Use the percent of sales method to forecast stockholders' equity.

10. What is the amount of net new financing needed for Global?

11. If Global decides that it will limit its net new financing to no more than $9 million, how will this affect its payout policy?

Context needed to solve it *The method these problems apply, from Berk §18.2, pp. 538–540 (PDF 569–571). Text, p. 538:* "In addition to the sales forecast, we require three other details to prepare the pro forma income statement: costs excluding depreciation in 2010 as a percent of sales, depreciation as a percent of sales, and the tax rate." *p. 539:* "The one final assumption we need to make is about our interest expense. We assume for now that it will remain the same as in 2010 because we will determine if our debt needs will change as part of the forecasting process." *p. 540:* "[…] our initial assumption is that debt will remain the same […]"

Table 18.2, "KMS Designs' Pro Forma Income Statement for 2011", p. 538 (PDF 569):

Income Statement ($000s)	2010	2011	Calculation
Sales	74,889	88,369	74,889 × 1.18
Costs Except Depreciation	−58,413	−68,928	78% of Sales
EBITDA	16,476	19,441	Lines 3 + 4
Depreciation	−5,492	−6,480	7.333% of Sales
EBIT	10,984	12,961	Lines 5 + 6
Interest Expense (net)	−306	−306	Remains the same
Pretax Income	10,678	12,655	Lines 7 + 8
Income Tax (35%)	−3,737	−4,429	35% of Line 9
Net Income	6,941	8,226	Lines 9 + 10

Table 18.3, "First-Pass Pro Forma Balance Sheet for 2011", p. 540 (PDF 571):

Balance Sheet ($000s)	2010	2011	Calculation
Cash and Cash Equivalents	11,982	14,139	16% of Sales

Balance Sheet ($000s)	2010	2011	Calculation
Accounts Receivable	14,229	16,790	19% of Sales
Inventories	14,978	17,674	20% of Sales
Total Current Assets	41,189	48,603	Lines 4 + 5 + 6
Property, Plant, and Equipment	49,427	58,324	66% of Sales
Total Assets	90,616	106,927	Lines 7 + 8
Accounts Payable	11,982	14,139	16% of Sales
Debt	4,500	4,500	Remains the same
Total Liabilities	16,482	18,639	Lines 11 + 12
Stockholders' Equity	74,134	79,892	74,134 + 70% of 8,226
Total Liabilities and Equity	90,616	98,531	Lines 13 + 14
Net New Financing		8,396	Line 9 − Line 15

Definition, p. 539 (PDF 570): Net New Financing = Projected Assets − Projected Liabilities and Equity.

---

9. KMS and its sustainable growth rate; new debt; D/E at the internal growth rate

Source: Berk, Ch. 18, Exercises 18.17–18.19, p. 561 (PDF 592) · Answer in book: no

17. Did KMS's expansion plan call for it to grow slower or faster than its sustainable growth rate?
18. Your firm has an ROE of 12%, a payout ratio of 25%, $600,000 of stockholders' equity, and $400,000 of debt. If you grow at your sustainable growth rate this year, how much additional debt will you need to issue?
19. IZAX, Co. had the following items on its balance sheet at the beginning of the year:

Assets		Liabilities and Equity	
Cash	50,000	Debt	100,000
PPE	350,000	Equity	300,000

Its net income this year is $20,000 and it pays dividends of $5,000. If its assets grew at its internal growth rate, what is its new D/E ratio?

Context needed to solve it *Formulas (18.4) and (18.5), Berk §18.4, p. 548 (PDF 579):* Internal Growth Rate = (Net Income/Beginning Assets) × (1 − Payout Ratio) = ROA × Retention Rate; Sustainable Growth Rate = (Net Income/Beginning Equity) × (1 − Payout Ratio) = ROE × Retention Rate. "[…] the firm's *sustainable growth rate*—the maximum growth rate the firm can sustain without issuing new equity or increasing its debt-to-equity ratio."

KMS's expansion plan (for Problem 17). Sales and net income from Table 18.8, "Pro Forma Income Statement for KMS Expansion", p. 545 (PDF 576); equity and dividends from Table 18.17, "Pro Forma Balance Sheet for KMS, 2010–2015", p. 562 (PDF 593). $000s:

Year	2010	2011	2012	2013	2014	2015
Sales (Table 18.8)	74,889	88,369	103,247	119,793	138,167	158,546
Net Income (Table 18.8)	6,941	7,600	8,807	11,141	13,739	16,627
Starting Stockholders' Equity (Table 18.17)	69,275	74,134	76,948	83,252	90,156	97,729
Net Income (Table 18.17)	6,940	7,600	8,807	11,141	13,739	16,627
Dividends (Table 18.17)	−2,082	−4,786	−2,503	−4,237	−6,167	−8,313
Stockholders' Equity (Table 18.17)	74,134	76,948	83,252	90,156	97,729	106,042
Debt (Table 18.17)	4,500	24,500	24,500	24,500	24,500	24,500

From Berk §18.3, p. 544 (PDF 575): KMS finances the $20 million of new equipment in 2011 with ten-year bonds; "KMS distributes all cash in excess of the minimum required cash as dividends" (p. 545).

10. Under Armour: sustainable growth and its sensitivity

Source: Higgins, Ch. 4, Exercise 4.8, pp. 135–136 (PDF 156–157) · Answer in book: no

8. Under Armour, Inc., is an American supplier of sportswear and casual apparel. Following are selected financial data for the company for the period 2009–2013.

	2009	2010	2011	2012	2013
Profit margin (%)	5.5	6.4	6.6	7.0	7.0
Retention ratio (%)	100.0	100.0	100.0	100.0	100.0
Asset turnover (X)	1.6	1.6	1.6	1.6	1.5
Financial leverage (X)	1.7	1.7	1.9	1.8	1.9
Growth rate in sales (%)	18.1	24.2	38.4	24.6	27.1

Source: Data from Under Armour 2009–2013 annual reports

a. Calculate Under Armour's annual sustainable growth rate for the years 2009 through 2013.

b. Did Under Armour face a growth management challenge during this period? Please explain briefly.

c. How did Under Armour cope with this challenge?

d. Calculate Under Armour's sustainable growth rate in 2013 assuming a profit margin of 8.0%. Return the profit margin to its original value and calculate the sustainable growth rate in 2013 assuming a financial leverage of 2.40 times. Calculate the sustainable growth rate in 2013 assuming both of these changes occur.

Context needed to solve it *From Higgins Ch. 4, "Sustainable Growth", pp. 113–114 (PDF 134–135):*

A company's sustainable growth rate therefore is nothing more than its growth rate in equity. Letting g^* represent the sustainable growth rate,

$$g^* = \frac{\text{Change in equity}}{\text{Equity}_{bop}}$$

where bop denotes beginning-of-period equity. Because the firm will not be selling any new shares by assumption, the only source of new equity will be from retained profits, so we can rewrite this expression as

$$g^* = \frac{R \times \text{Earnings}}{\text{Equity}_{bop}}$$

where R is the firm's "retention rate." […] The ratio "Earnings/Equity" in this expression should look familiar; it is the firm's return on equity, or ROE. Thus, $g^* = R \times \text{ROE}_{bop}$. Finally, recalling the levers of performance discussed in Chapter 2, we can rewrite this expression yet again as

$$g^* = PRA\hat{T}$$

where P, A, and $\hat{T}$ are our old friends from Chapter 2, the levers of performance, where P is the profit margin, A is the asset turnover ratio, and $\hat{T}$ is the assets-to-equity ratio. The assets-to-equity ratio wears a hat here as a reminder that it is assets divided by beginning-of-period equity instead of end-of-period equity as defined in Chapter 2. […] If a company increases sales at any rate other than g^*, one or more of the ratios must change. This means that when a company grows at a rate in excess of its sustainable growth rate, it had better improve operations (represented by an increase in the profit margin or the asset turnover ratio) or prepare to alter its financial policies (represented by increasing its retention rate or its financial leverage).

11. Turismo da Onça: NAF, excess capacity, constant debt-equity

Source: Ross, Cap. 4, Exercises 4.25–4.27, p. 121 (PDF 145) · Answer in book: no (selected answers in the web-only Apêndice C)

25. **Cálculo da NAF** [OA2] A seguir estão as demonstrações contábeis mais recentes da Turismo da Onça S/A. As vendas de 2009 têm crescimento projetado de 20%. As despesas com juros, a alíquota tributária e a taxa de distribuição de dividendos permanecerão constantes. Os custos, as outras despesas, o ativo circulante e as contas a pagar aumentam na proporção das vendas. Se a empresa opera à capacidade máxima e não foram emitidas ações ou assumidas novas dívidas, qual é o aporte financeiro necessário para suportar a taxa de crescimento de 20% nas vendas?

TURISMO DA ONÇA S/A — Demonstração de resultados de 2008

Vendas		$929.000
Custos		723.000
Outras despesas		19.000
Lucros antes de juros e impostos		$187.000
Juros pagos		14.000
Lucro tributável		$173.000
Impostos		60.550
Lucro líquido		$112.450
Dividendos	$33.735	
Acréscimo aos lucros retidos	78.715	

TURISMO DA ONÇA S/A — Balanço patrimonial de 31 de Dezembro de 2008

Ativo		Passivo e patrimônio líquido	
Ativo circulante		*Passivo circulante*	
Caixa	$ 25.300	Contas a pagar	$ 68.000
Contas a receber	40.700	Empréstimos a pagar	17.000
Estoque	86.900	Total	$ 85.000
Total	$152.900	Passivo não circulante	$158.000
Ativo imobilizado		*Patrimônio líquido*	
Instalações e equipamentos	413.000	Capital social e ágio	$140.000
		Lucros retidos	182.900
		Total	$322.900
Total do ativo	$565.900	Total do passivo e do patrimônio líquido	$565.900

26. **Uso da capacidade e crescimento** [OA2] No problema anterior, suponha que a empresa tenha operado a apenas 80% da capacidade em 2008. Qual é a NAF agora?

27. **Cálculo da NAF** [OA2] No Problema 25, suponha que a empresa queira manter seu índice dívida/capital próprio constante. Qual é a NAF agora?

Context needed to solve it *From Ross §4.3, "Uma situação alternativa" (capacity), p. 104 (PDF 128):*

Se assumirmos que a Mar de Rosas está operando a apenas 70% de sua capacidade, a necessidade de aportes financeiros será bastante diferente. Quando dizemos "70% da capacidade", queremos dizer que o nível atual de vendas é 70% da capacidade total de vendas:

Vendas atuais = $1.000 = 0,70 × Capacidade total de vendas Capacidade total de vendas = $1.000/0,70 = $1.429.

Isso nos mostra que as vendas poderiam aumentar em quase 43% – de $1.000 para $1.429 – antes que fossem necessários novos ativos imobilizados.

Tier 3 — Really hard

12. R&E Supplies: extending the spreadsheet forecast, sensitivity and scenario analysis

Source: Higgins, Ch. 3, Exercise 3.9, p. 105 (PDF 126); Part 0 = Exercise 3.8, p. 105 (PDF 126) · Answer in book: yes for 3.9, p. 406; no for 3.8

9. A spreadsheet containing R&E Supplies' 2022 pro forma financial forecast, as shown in Table 3.5, is available for download from McGraw-Hill's *Connect* or your course instructor (see the Preface for more information). Using the spreadsheet information presented next, and the modified equations determined in Problem 8 earlier, extend the forecast for R&E Supplies contained in Table 3.5 through 2023.

R&E Supplies Assumptions for 2023 ($ thousands)

Growth rate in net sales	30.0%	Tax rate	25.0%
Cost of goods sold/net sales	86.0%	Dividend/earnings after tax	50.0%
Gen., sell., & admin. expenses/net sales	11.0%	Current assets/net sales	29.0%
Long-term debt	$560	Net fixed assets	$ 270
Current portion long-term debt	$100	Current liabilities/net sales	14.4%
Interest rate	10.0%		

a. What is R&E's projected external financing required in 2023? How does this number compare to the 2022 projection?

b. Perform a sensitivity analysis on this projection. How does R&E's projected external financing required change if the ratio of cost of goods sold to net sales declines from 86.0 percent to 84.0 percent?

c. Perform a scenario analysis on this projection. How does R&E's projected external financing required change if a severe recession occurs in 2023? Assume net sales decline 5 percent, cost of goods sold rises to 88 percent of net sales due to price cutting, and current assets increase to 35 percent of net sales as management fails to cut purchases promptly in response to declining sales.

Context needed to solve it **Part 0** — *Source: Higgins, Ch. 3, Exercise 3.8, p. 105 (PDF 126) · Answer in book: no*

8. Table 3.5 presents a computer spreadsheet for estimating R&E Supplies' external financing required for 2022. The text mentions that with modifications to the equations for equity and net sales, the forecast can easily be extended through 2023. By referencing the appropriate cells, write the modified equations for equity and net sales for 2023.

Table 3.5, "Forecasting with a Spreadsheet: Pro Forma Financial Forecast for R&E Supplies, Inc., December 31, 2022 ($ thousands)", p. 90 (PDF 111). Column A = labels, B = 2021 actual (rows 3–15) or the equations (rows 19–40), C = 2022, D = 2023 (blank):

Row	A	B	C	D
2	Year	2021 Actual	2022	2023
3	Net sales	$20,613		
4	Growth rate in net sales		25.0%	
5	Cost of goods sold/net sales		86.0%	
6	Gen., sell., and admin. expenses/net sales		12.0%	
7	Long-term debt	$ 760	$660	
8	Current portion long-term debt	$ 100	$100	
9	Interest rate		10.0%	
10	Tax rate		25.0%	
11	Dividend/earnings after tax		50.0%	
12	Current assets/net sales		29.0%	
13	Net fixed assets		$280	
14	Current liabilities/net sales		14.5%	
15	Owners' equity	$ 1,730		

Row	A	B	C	D
16	INCOME STATEMENT			
17		Equations	Forecast	Forecast
18	Year	2022	2022	2023
19	Net sales	= B3 + B3 * C4	$25,766	
20	Cost of goods sold	= C5*C19	22,159	
21	Gross profit	= C19 – C20	3,607	
22	Gen., sell., and admin. exp.	= C6*C19	3,092	
23	Interest expense	= C9*(C7 + C8 + C40)	228	
24	Earnings before tax	= C21 – C22 – C23	287	
25	Tax	= C10*C24	72	
26	Earnings after tax	= C24 – C25	216	
27	Dividends paid	= C11*C26	108	
28	Additions to retained earnings	= C26 – C27	108	
30	BALANCE SHEET			
31	Current assets	= C12*C19	7,472	
32	Net fixed assets	= C13	280	
33	Total assets	= C31 + C32	7,752	
35	Current liabilities	= C14*C19	3,736	
36	Long-term debt	= C7	660	
37	Owners' equity	= B15 + C28	1,838	
38	Total liabilities and owners' equity	= C35 + C36 + C37	6,234	
40	EXTERNAL FUNDING REQUIRED	= C33 – C38	$ 1,518	

From Higgins Ch. 3, "Forecasting with Spreadsheets", pp. 89–92 (PDF 110–113):

The first area on the simulated screen is an assumptions box, containing all of the information and assumptions required to construct the forecast. […] The forecast begins immediately below the assumptions box. The first column, labeled "Equations 2022," is included for explanatory purposes and would not appear on a conventional forecast. Entering the equations shown causes the computer to calculate the quantities appearing in the second column, labeled "Forecast 2022." The third column, labeled "Forecast 2023," is presently blank. […] The first equation for net sales is = B3 + B3 * C4. This instructs the spreadsheet to get the number in cell B3 and add to it that number times the number in cell C4—in other words, $20,613 + $20,613 × 25%. […] There are only three tricky equations. Interest expense, row 23, is the interest rate times end-of-period long-term debt, including the current portion, plus the forecasted external funding required. As discussed earlier, the tricky part here is the interdependence between interest expense and external funding required. […] The equity equation, row 37, is end-of-period equity plus additions to retained earnings; the external funding required equation, row 40, is total assets minus total liabilities and owners' equity.

---

13. Deriving NAF and the growth-rate formulas

Source: Ross, Cap. 4, Exercises 4.31–4.33 (DESAFIO), p. 122 (PDF 146) · Answer in book: no (selected answers in the web-only Apêndice C)

31. **NAF** [OA2] Defina o seguinte:

V = Vendas do ano anterior A = Total do ativo D = Dívida total P = Total do patrimônio líquido g = Projeção de crescimento das vendas ML = Margem de lucro b = Taxa de retenção

Mostre que a NAF pode ser escrita da seguinte forma:

$$\text{NAF} = -\text{ML}(V)b + (A - \text{ML}(V)b) \times g$$

Dica: as necessidades de ativo serão iguais a $A \times g$. O acréscimo aos lucros retidos será igual a $\text{ML}(V)b \times (1 + g)$.

32. **Taxas de crescimento** [OA2] Com base no resultado do Problema 31, mostre que as taxas de crescimento interna e sustentável estão de acordo com aquelas mostradas no capítulo. Dica: para a taxa de crescimento interna, defina NAF igual a zero e calcule g.

33. **Taxa de crescimento sustentável** [OA3] Neste capítulo, discutimos as duas versões da fórmula da taxa de crescimento sustentável. Derive a fórmula ROE $\times$ b a partir da fórmula dada no capítulo, onde o ROE se baseia no patrimônio líquido inicial. Da mesma maneira, derive a fórmula ROA $\times$ b a partir da fórmula da taxa de crescimento interna.

Context needed to solve it *From Ross §4.4, "Política financeira e crescimento", p. 109 (PDF 133), and Tabela 4.9, p. 111 (PDF 135):*

$$\text{Taxa de crescimento interna} = \frac{\text{ROA} \times b}{1 - \text{ROA} \times b} \qquad [4.2]$$

onde o ROA é o retorno sobre o ativo discutido no Capítulo 3, e b é a taxa de retenção discutida anteriormente neste capítulo. [...]

$$\text{Taxa de crescimento sustentável} = \frac{\text{ROE} \times b}{1 - \text{ROE} \times b} \qquad [4.3]$$

Isso é idêntico à taxa de crescimento interna, exceto que é usado o retorno sobre o patrimônio líquido (ROE) em vez do ROA.

Tabela 4.9: ROA = Retorno sobre o ativo = Lucro líquido / Total do ativo; ROE = Retorno sobre o patrimônio líquido = Lucro líquido/Total do patrimônio líquido; b = Taxa de retenção = Acréscimo aos lucros retidos/Lucro líquido. "A taxa de crescimento interna é a taxa máxima de crescimento que pode ser atingida sem qualquer aporte financeiro." "A taxa de crescimento sustentável é a taxa máxima de crescimento que uma empresa pode atingir sem precisar emitir novas ações, enquanto mantém um índice dívida/capital próprio constante."

From Ross, "Uma observação sobre os cálculos da taxa de crescimento sustentável", p. 112 (PDF 136):

Usualmente, a taxa de crescimento sustentável é calculada utilizando apenas o numerador de nossa equação, (ROE $\times$ b). Isso causa alguma confusão, que podemos esclarecer aqui. A questão relaciona-se com como o ROE é calculado. lembre-se de que ele é calculado pela divisão do lucro líquido pelo total do patrimônio líquido. Caso o total do patrimônio líquido seja tomado dos números de fim de período do balanço patrimonial (como temos feito até aqui e, em geral, é feito na prática), nossa fórmula está correta. Entretanto, se o total do patrimônio líquido corresponde ao início do período, então a fórmula simplificada será a correta.

---

14. Turismo da Onça: NAF as a function of growth

Source: Ross, Cap. 4, Exercises 4.28 and 4.29 (DESAFIO), p. 121 (PDF 145); Part 0 = Exercises 4.25 and 4.27, p. 121 (PDF 145) · Answer in book: no (selected answers in the web-only Apêndice C)

28. **NAF e crescimento interno** [OA2, OA3] Refaça o Problema 25 usando agora, além da taxa de crescimento de vendas de 20%, as taxas de 15 e 25%. Ilustre graficamente a relação entre a NAF e a taxa de crescimento. A qual taxa de crescimento a NAF é igual a zero? Por que essa taxa de crescimento interna é diferente daquela encontrada com o uso da equação no texto?

29. **NAF e crescimento interno** [OA2, OA3] Refaça o Problema 27 usando taxas de crescimento de vendas de 30 e 35%, além da de 20%. Ilustre graficamente a relação entre a NAF e a taxa de crescimento. A qual taxa de crescimento a NAF é igual a zero? Por que essa taxa de crescimento sustentável é diferente daquela que encontramos usando a equação no texto?

Context needed to solve it **Part 0** — *Source: Ross, Cap. 4, Exercises 4.25 and 4.27, p. 121 (PDF 145) · Answer in book: no (these are also exercise 11 of this list)*

25. **Cálculo da NAF** [OA2] A seguir estão as demonstrações contábeis mais recentes da Turismo da Onça S/A. As vendas de 2009 têm crescimento projetado de 20%. As despesas com juros, a alíquota tributária e a taxa de distribuição de dividendos permanecerão constantes. Os custos, as outras despesas, o ativo circulante e as contas a pagar aumentam na proporção das vendas. Se a empresa opera à capacidade máxima e não foram emitidas ações ou assumidas novas dívidas, qual é o aporte financeiro necessário para suportar a taxa de crescimento de 20% nas vendas?

TURISMO DA ONÇA S/A — Demonstração de resultados de 2008

Vendas		$929.000
Custos		723.000
Outras despesas		19.000
Lucros antes de juros e impostos		$187.000
Juros pagos		14.000
Lucro tributável		$173.000
Impostos		60.550
Lucro líquido		$112.450
Dividendos	$33.735	
Acréscimo aos lucros retidos	78.715	

TURISMO DA ONÇA S/A — Balanço patrimonial de 31 de Dezembro de 2008

Ativo		Passivo e patrimônio líquido	
Ativo circulante		*Passivo circulante*	
Caixa	$ 25.300	Contas a pagar	$ 68.000
Contas a receber	40.700	Empréstimos a pagar	17.000
Estoque	86.900	Total	$ 85.000
Total	$152.900	Passivo não circulante	$158.000
Ativo imobilizado		*Patrimônio líquido*	
Instalações e equipamentos	413.000	Capital social e ágio	$140.000
		Lucros retidos	182.900
		Total	$322.900
Total do ativo	$565.900	Total do passivo e do patrimônio líquido	$565.900

27. **Cálculo da NAF** [OA2] No Problema 25, suponha que a empresa queira manter seu índice dívida/capital próprio constante. Qual é a NAF agora?

"A equação no texto": Ross §4.4, p. 109 (PDF 133), and Tabela 4.9, p. 111 (PDF 135):

$$\text{Taxa de crescimento interna} = \frac{\text{ROA} \times b}{1 - \text{ROA} \times b} \qquad [4.2] \qquad\qquad \text{Taxa de crescimento sustentável} = \frac{\text{ROE} \times b}{1 - \text{ROE} \times b} \qquad [4.3]$$

onde ROA = Lucro líquido / Total do ativo; ROE = Lucro líquido/Total do patrimônio líquido; b = Taxa de retenção = Acréscimo aos lucros retidos/Lucro líquido.

---

15. KMS Designs: valuing the expansion when market share grows 0.25% a year

Source: Berk, Ch. 18, Exercises 18.20–18.22, p. 561 (PDF 592); Part 0 = Exercises 18.12–18.15, pp. 560–561 (PDF 591–592) · Answer in book: no

20. Forecast KMS's free cash flows (reproduce Table 18.13), assuming KMS's market share will increase by 0.25% per year; investment, financing, and depreciation will be adjusted accordingly; and working capital will be as you projected in Problem 15).

21. Calculate the continuation value of KMS using your reproduction of Table 18.8 from Problem 14, and assuming an EBITDA multiple of 8.5.

22. Assuming a cost of capital of 10%, compute the value of KMS under the 0.25% growth scenario.

Context needed to solve it Part 0 — *Source: Berk, Ch. 18, Exercises 18.12–18.15, pp. 560–561 (PDF 591–592) · Answer in book: no.* (The book adds "For problems in this section, you should download the KMS spreadsheets available on the textbook's Web site"; the chapter tables copied below contain every number those spreadsheets hold.)

12. Assume that KMS's market share will increase by 0.25% per year rather than the 1% used in the chapter (see Table 18.5) and that its prices remain as in the chapter. What production capacity will KMS require each year? When will an expansion become necessary (that is, when will production volume exceed 1100)?

13. Under the assumption that KMS's market share will increase by 0.25% per year, you determine that the plant will require an expansion in 2012. The expansion will cost $20 million. Assuming that the financing of the expansion will be delayed accordingly, calculate the projected interest payments and the amount of the projected interest tax shields (assuming that KMS still uses a ten-year bond and interest rates remain the same as in the chapter) through 2015.

14. Under the assumption that KMS's market share will increase by 0.25% per year (and the investment and financing will be adjusted as described in Problem 13), you project the following depreciation:

Year	2010	2011	2012	2013	2014	2015
Depreciation	5,492	5,443	7,398	7,459	7,513	7,561

Using this information, project net income through 2015 (that is, reproduce Table 18.8 under the new assumptions).

15. Assuming that KMS's market share will increase by 0.25% per year (implying that the investment, financing, and depreciation will be adjusted as described in Problems 13 and 14), and that the working capital assumptions used in the chapter still hold, calculate KMS's working capital requirements through 2015 (that is, reproduce Table 18.9 under the new assumptions).

The chapter's KMS model, Berk §18.3–18.5, pp. 543–555 (PDF 574–586).

p. 543: KMS currently has the capacity to produce a maximum of 1.1 million units (i.e., 1100 thousand units). However, as detailed in Table 18.5, KMS expects both the total market size and its share of the market to grow to the point where the company will quickly exceed that capacity. Thus, KMS is considering an expansion that will increase its capacity to 2 million units—enough to handle its projected requirements through 2015.

Table 18.5, "KMS's Forecasted Production Capacity Requirements", p. 543:

Year	2010	2011	2012	2013	2014	2015
Market Size (000s units)	10,000	10,500	11,025	11,576	12,155	12,763
Market Share	10.0%	11.0%	12.0%	13.0%	14.0%	15.0%
Production Volume (Line 3 × Line 4)	1,000	1,155	1,323	1,505	1,702	1,914
Average Sales Price	$ 74.89	$ 76.51	$ 78.04	$ 79.60	$ 81.19	$ 82.82

pp. 543–544: The new equipment to increase KMS's capacity will cost $20 million and will need to be purchased in 2011 to meet the company's production needs. […] KMS has ongoing capital investment requirements to cover the replacement of existing equipment—these were expected to be $5 million per year without the new equipment. The additional $20 million is reflected in 2011, bringing the total to $25 million for 2011 and increasing expected recurring investment to $8 million per year in years 2012–2015.

Table 18.6, "KMS's Forecasted Capital Expenditures" ($000s), p. 543:

Year	2010	2011	2012	2013	2014	2015
Opening Book Value	49,919	49,427	66,984	67,486	67,937	68,344
Capital Investment	5,000	25,000	8,000	8,000	8,000	8,000
Depreciation	−5,492	−7,443	−7,498	−7,549	−7,594	−7,634
Closing Book Value	49,427	66,984	67,486	67,937	68,344	68,709

p. 544: KMS plans to finance the new equipment by issuing ten-year coupon bonds with a coupon rate of 6.8%. Thus, KMS will pay only interest on the bonds until the repayment of principal in ten years. The principal on its outstanding debt of $4500 is also not due before 2015.

Table 18.7, "KMS's Planned Debt and Interest Payments" ($000s), p. 544 (rate 6.80%):

Year	2010	2011	2012	2013	2014	2015
Outstanding Debt	4,500	24,500	24,500	24,500	24,500	24,500
Net New Borrowing	—	20,000	—	—	—	—
Interest on Debt	306	306	1,666	1,666	1,666	1,666

$$\text{Interest in Year } t = \text{Interest Rate} \times \text{Ending Balance in Year } (t-1) \qquad (18.1)$$

$$\text{Sales} = \text{Market Size} \times \text{Market Share} \times \text{Average Sales Price} \qquad (18.2)$$

p. 544: We will assume that costs except depreciation will continue to be 78% of sales […]. *p. 545:* KMS pays a 35% tax rate, and the income tax is computed as: Income Tax = Pretax Income × Tax Rate (18.3).

Table 18.8, "Pro Forma Income Statement for KMS Expansion" ($000s), p. 545:

Year	2010	2011	2012	2013	2014	2015
Sales	74,889	88,369	103,247	119,793	138,167	158,546
Costs Except Depreciation	−58,413	−68,928	−80,533	−93,438	−107,770	−123,666
EBITDA	16,476	19,441	22,714	26,354	30,397	34,880
Depreciation	−5,492	−7,443	−7,498	−7,549	−7,594	−7,634
EBIT	10,984	11,998	15,216	18,806	22,803	27,246
Interest Expense (net)	−306	−306	−1,666	−1,666	−1,666	−1,666
Pretax Income	10,678	11,692	13,550	17,140	21,137	25,580
Income Tax	−3,737	−4,092	−4,742	−5,999	−7,398	−8,953
Net Income	6,941	7,600	8,807	11,141	13,739	16,627

p. 545: We have forecast that the minimum required cash will be 16% of sales, accounts receivable will be 19% of sales, inventory will be 20% of sales, and accounts payable will be 16% of sales, all as they were in 2010.

Table 18.9, "KMS Projected Working Capital Needs" ($000s), p. 545:

Year	2010	2011	2012	2013	2014	2015
Cash	11,982	14,139	16,520	19,167	22,107	25,367
Accounts Receivable	14,229	16,790	19,617	22,761	26,252	30,124
Inventory	14,978	17,674	20,649	23,959	27,633	31,709
Total Current Assets	41,189	48,603	56,786	65,886	75,992	87,201
Accounts Payable	11,982	14,139	16,520	19,167	22,107	25,367
Total Current Liabilities	11,982	14,139	16,520	19,167	22,107	25,367
Net Working Capital (7 − 10)	29,207	34,464	40,266	46,719	53,885	61,833

Year	2010	2011	2012	2013	2014	2015
Increase in Net Working Capital		5,257	5,802	6,453	7,166	7,948

pp. 551–552 (PDF 582–583): To compute KMS's free cash flows, which exclude cash flows associated with leverage, we first adjust net income by adding back the after-tax interest payments associated with the net debt in its capital structure. […] After-Tax Interest Expense = (1 − Tax Rate) × (Interest on Debt) − (1 − Tax Rate) × (Interest on Excess Cash) (18.6). […] To compute KMS's free cash flows from its unlevered net income, we add back depreciation (which is not a cash expense), and deduct KMS's increases in net working capital and capital expenditures.

Table 18.13, "KMS Forecasted Free Cash Flows" ($000s), p. 551:

Year	2011	2012	2013	2014	2015
Net Income	7,600	8,807	11,141	13,739	16,627
Plus: After-Tax Interest Expense	199	1,083	1,083	1,083	1,083
Unlevered Net Income	7,799	9,890	12,224	14,822	17,710
Plus: Depreciation	7,443	7,498	7,549	7,594	7,634
Less: Increases in NWC	(5,257)	(5,802)	(6,453)	(7,166)	(7,948)
Less: Capital Expenditures	(25,000)	(8,000)	(8,000)	(8,000)	(8,000)
Free Cash Flow of Firm	(15,015)	3,586	5,320	7,250	9,396

pp. 553–555 (PDF 584–586): As we learned in Chapter 16, absent distress costs, the value of a firm with debt is equal to the value of the firm without debt plus the present value of its interest tax shields. […] we compute the present value of the unlevered free cash flows of KMS and add to it the present value of the tax shields created by our planned interest payments. […] Continuation Enterprise Value at Forecast Horizon = EBITDA at Horizon × EBITDA Multiple at Horizon (18.7). […] Assume that KMS's financial managers have estimated KMS's unlevered cost of capital to be 10% (specifically, 10% is their pretax WACC […]). […] we discount KMS's free cash flows at the firm's pretax WACC of 10%. […] Interest Tax Shield = Net Interest Expense × Tax Rate (18.10). […] we calculate the present value of the interest tax shield using the interest rate on debt as the discount rate, not using the WACC. Recall that the reason for doing so is that the tax shield is only as risky as the debt that creates it, so that the proper discount rate is the debt's interest rate, here at 6.8% […]. Footnote 14: We have not ignored the rest of the interest tax shields from the new debt. The value of those shields is subsumed in the continuation value of the firm.

Table 18.14, "Calculation of KMS Firm Value with the Expansion" ($000s), p. 554 (PDF 585):

Year	2010	2011	2012	2013	2014	2015
Free Cash Flow of Firm		(15,015)	3,586	5,320	7,250	9,396
PV Free Cash Flow (at 10%)	4,096					
Continuation Value						313,920
PV Continuation Value (at 10%)	194,920					
Net Interest Expense		306	1,666	1,666	1,666	1,666
Interest Tax Shield		107	583	583	583	583
PV Interest Tax Shield (at 6.8%)	1,958					
Firm Value (3 + 5 + 8)	200,974					

(In the chapter, 2015 EBITDA of $34.880 million × a multiple of 9 = $313.920 million.)